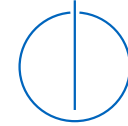


Robotics, Artificial Intelligence and Embedded Systems



Cognitive Systems: Neural Networks and Applied AI

Part I: Foundations and Theory of Cognitive Systems

Chapter 2

Cognitive Systems – A Definition

Prof. Dr.-Ing. habil. Alois Knoll

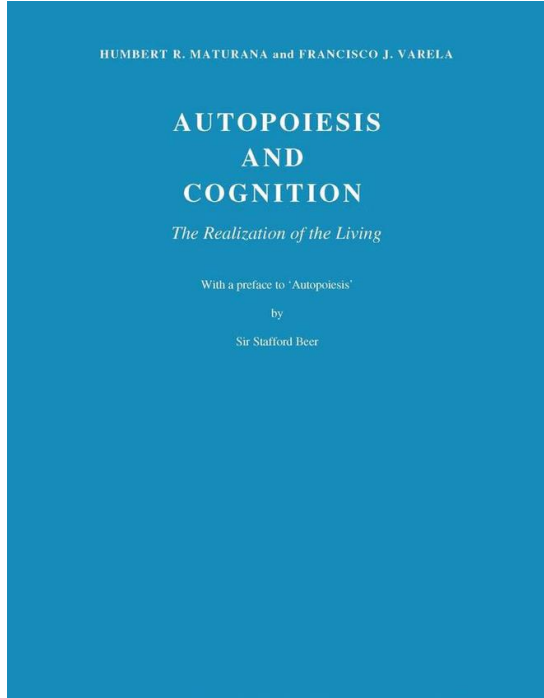
Florian Walter, M.Sc.

Topics in this Chapter

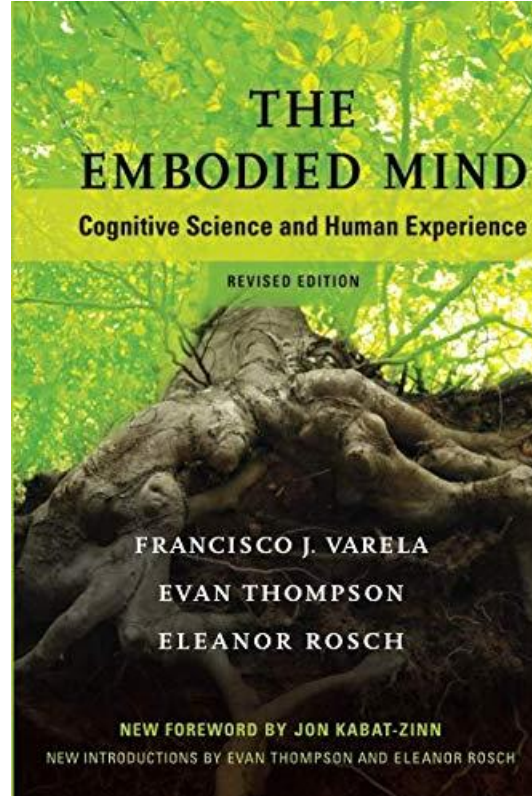
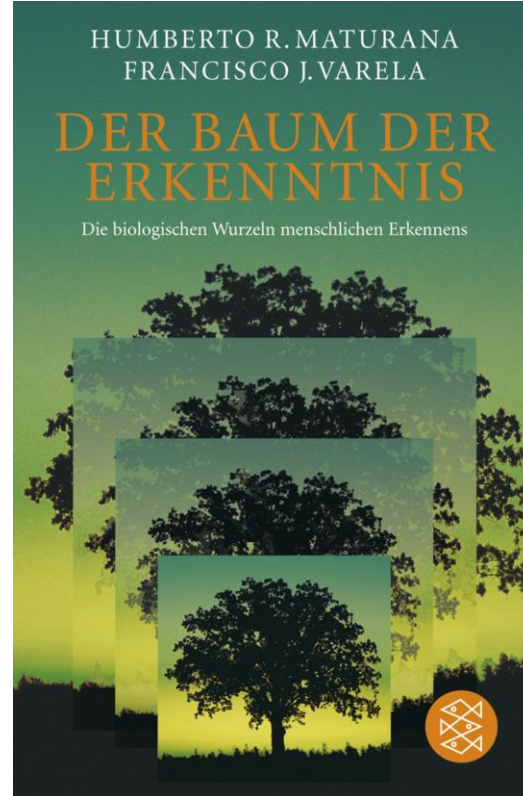
*From cognition...
... to cognitive systems*

- What is cognition and what is it good for?
- How is cognition related to intelligence and learning?
- What are the main components of a cognitive system?
- What are cognitive systems capable of?
- Is it possible to measure cognition?

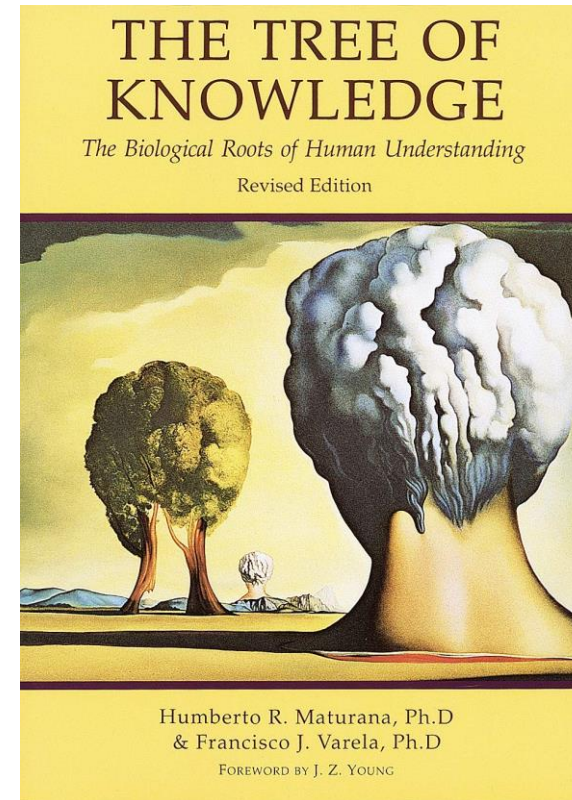
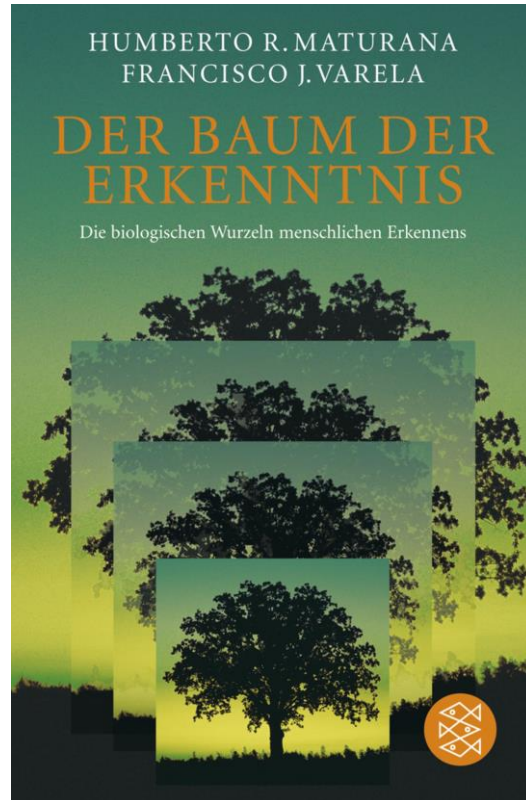
Key Literature



 **kluwer**
the language of science



On the Importance of Language



Simple direct translations in most cases do not preserve the actual meaning!

Der Baum der Erkenntnis (Gen 2,9)



Hauptseite
Themenportale
Zufälliger Artikel

Mitmachen
Artikel verbessern
Neuen Artikel anlegen
Autorenportal
Hilfe
Letzte Änderungen
Kontakt
Spenden

Werkzeuge
Links auf diese Seite
Änderungen an verlinkten Seiten
Spezialseiten
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Seiteninformationen
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Artikel zitieren

Nicht angemeldet Diskussionssseite Beiträge Benutzerkonto erstellen Anmelden

Artikel Diskussion Lesen Bearbeiten Mehr

Baum der Erkenntnis

Dieser Artikel behandelt den biblischen Baum. Zum philosophischen Werk siehe **Der Baum der Erkenntnis**.

Der **Baum der Erkenntnis** von *Gut und Böse* (**hebr.** עץ הדעת טוב ורע *ez had-da'at tôb wâ-râ*, **griech.** τὸ ξύλον τοῦ εἰδέναι γνωστὸν καλοῦ καὶ πονηροῦ, **lat.** *lignum sapientiae boni et mali*) ist ein Baum in der Paradieserzählung des Buches **Genesis** der **Bibel**. Er befindet sich zusammen mit dem **Baum des Lebens** in der Mitte des Paradiesgartens (**Garten Eden**) (**Gen** 2,9 [EU](#)^[?]). Gott verbot den Menschen, von dessen Früchten zu essen (**Gen** 2,17 [EU](#)^[?]).

Inhaltsverzeichnis [Verbergen]

- Von der historischen zur symbolischen Deutung
- Deutungszusammenhang von Urgeschichte und Eschatologie
 - Verlust der ursprünglichen



Lucas Cranach d. Ä.: Der Baum der Erkenntnis, Ausschnitt des Gemäldes „Paradies“ im Kunsthistorischen Museum, Wien



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Tree of the knowledge of good and evil

From Wikipedia, the free encyclopedia

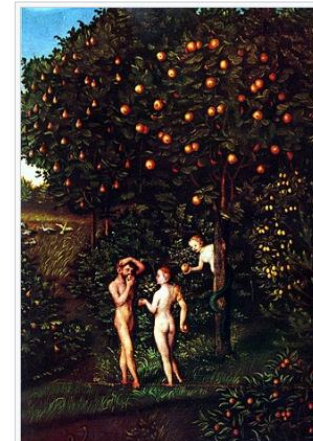
*For other uses, see **Tree of Knowledge** (disambiguation).*

The **tree of the knowledge of good and evil** (Biblical Hebrew: עֵץ הַדַּעַת טוֹב וָרָע *ez had-da'at ṭov wā-rā* [ˈes̺əd̺aʕaθ̺ t̺ov wɑ̺rɑʕ̺]) is one of two specific trees in the story of the **Garden of Eden** in Genesis 2–3, along with the **tree of life**.

Contents [hide]

- In Genesis
 - Narrative
 - Meaning of *good and evil*
- Religious views
 - Judaism
 - Christianity
 - Gnosticism

knowledge:
Kenntnis, Wissen, Erkenntnis,
Wissensstand



"The Fall of Man" by Lucas Cranach the Elder. The Tree of Knowledge is on the right.

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Artículo Discusión Leer Editar Ver historial

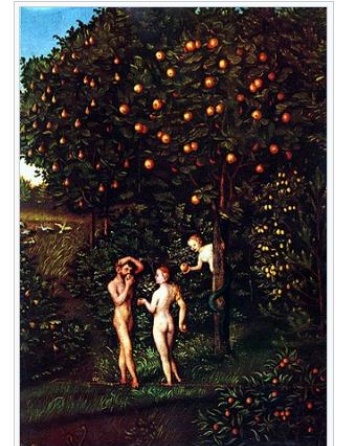
Árbol del conocimiento del bien y del mal

El **árbol del conocimiento del bien y el mal**¹ (en **hebreo**, עץ הדעת טוב ורע *Etz ha-da'at tov va-ra*) es uno de los dos **Árboles del Paraíso** en la historia del **Jardín del Edén**, en Génesis 2-3, junto con el **árbol de la vida**.

Índice [ocultar]

- En Génesis
 - Motivo
 - Composición
- Visión religiosa
 - Judaísmo
 - Cristianismo
 - Islam
 - Otras culturas
 - Etnomicología
- Véase también

el conocimiento:
Wissen, Kenntnis, Kenntnisnahme, Mitwissen, Einsicht



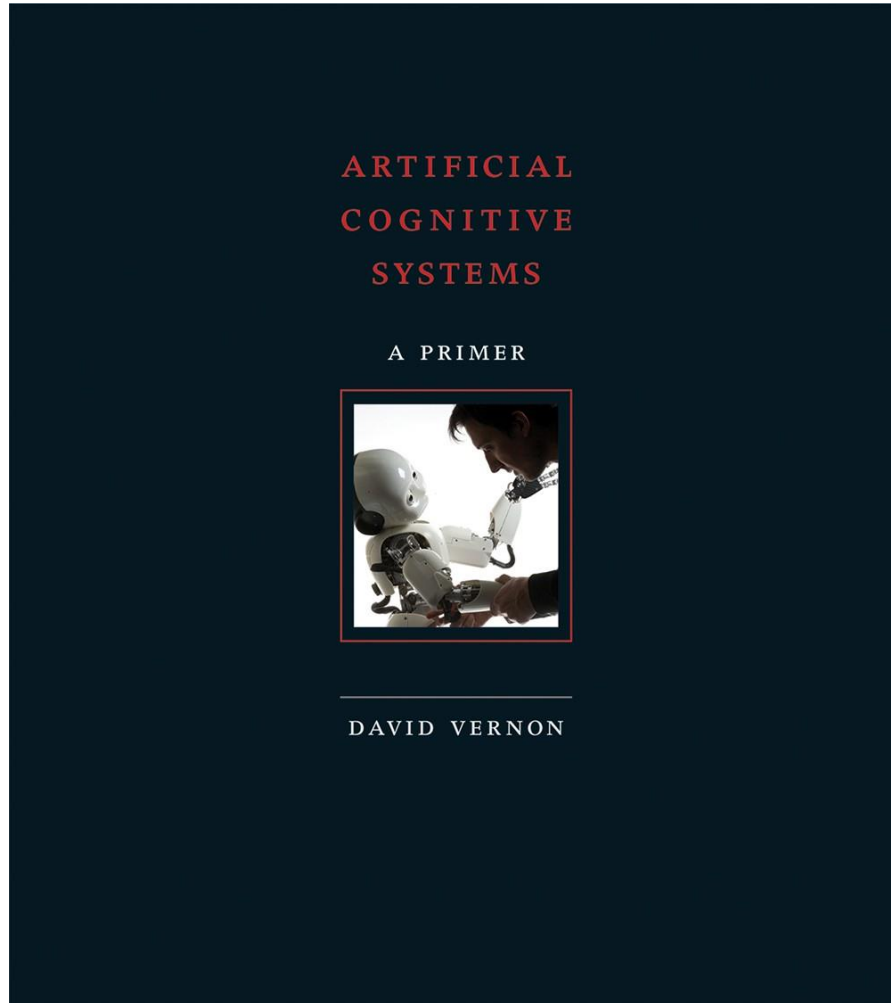
La caída del hombre, de Lucas Cranach el Viejo. El árbol del conocimiento aparece a la derecha.

Defining and Describing Cognitive Systems

- Discussions about cognitive systems frequently touch upon complex philosophical terms and concepts
- Simple direct translations do not capture the meaning of the original text

This course is held bilingually with English slides and German presentation to fully convey the ideas behind the presented concepts

Introduction and Reference

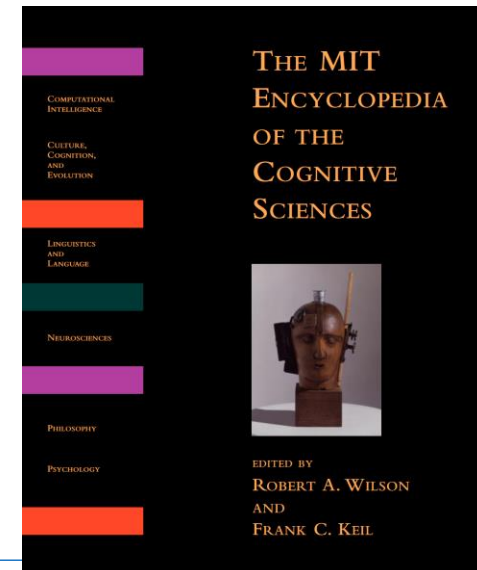


Vernon, David (2014): [Artificial cognitive systems. A primer](#). Cambridge, MA: The MIT Press.

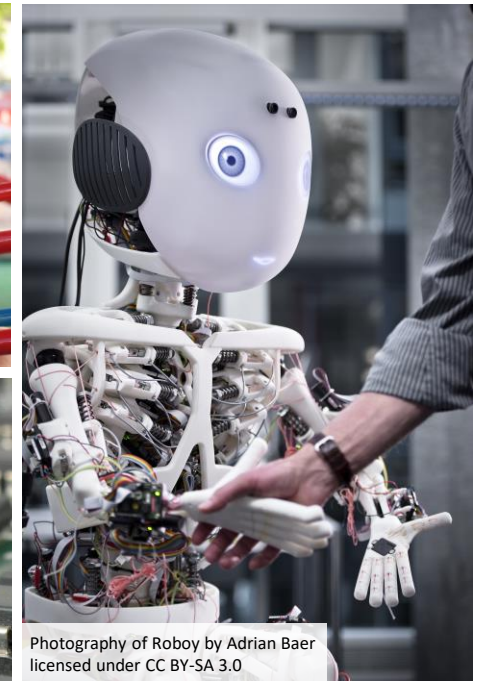
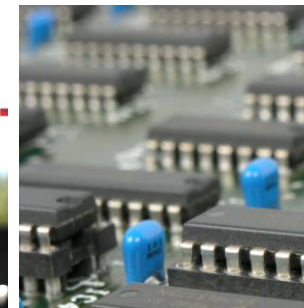
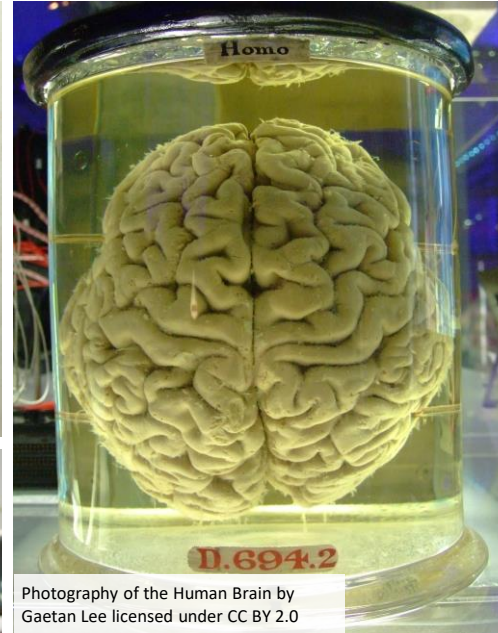
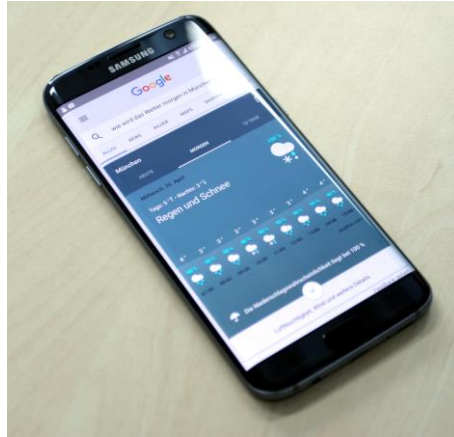
Slides by the author of the book are available on

<http://www.vernon.eu/courses.htm#cogsys>

<http://www.aiai.ed.ac.uk/project/oplan/documents/1999/1999-MITECS.pdf>



What is a Cognitive System?



Cognition

Cognition, lat. *cognitio*

The **process** involved in **knowing**, or the act of knowing, which in its completeness includes **perception** and **judgment**. Cognition includes all processes of **consciousness** by which knowledge is accumulated, such as **perceiving**, **recognizing**, **conceiving**, and **reasoning**. Put differently, cognition is an **experience of knowing** that can be distinguished from an experience of feeling or willing. It is one of the only words that refers to the **brain** as well as to the **mind**.

From Encyclopædia Britannica

Kognition

Gesamtheit aller Prozesse, die mit dem Wahrnehmen und Erkennen zusammenhängen

Duden

System

System, grk. *systema*

A regularly interacting or interdependent **group** of items forming a **unified whole**:

- A group of **interacting** bodies under the influence of related **forces**
- A group of body organs that **together** perform one or more vital functions
- The body considered as a **functional unit**
- A group of **devices** or **artificial objects** or an organization forming a **network** especially for **distributing** something or serving a **common purpose**

Adapted from Merriam Webster

The Purpose of Cognition

Cognition as a Mechanism of Active, Self-Reliant, and Autonomous Behavior

- Cognition is the **sum of all mechanisms** that enable a system to deal with the uncertainty and change within real-world environments
- The maintenance of a well-defined system state even under hostile conditions is made possible by fixed **goals** and **intentions** that guide the **decisions** and **actions** of the system in a purposeful way
- Cognition enables **interaction** with other cognitive agents by anticipating and respecting their individual goals
- Cognition is an essential prerequisite for a system's **autonomy**

Cognition enables a system to operate in a meaningful way beyond its original pre-programmed behavior and specification!

The Purpose of Cognition

The Complexity of “Simple Tasks”

- Technical systems are based on and operate according to exact **formal descriptions, rules** and **specifications**
- The design and implementation of these descriptions is one of the main research questions in computer science (→ **Software Engineering**)
- Seemingly **simple real-world tasks** from everyday life cannot be described with formal methods – there is no formal “specification” of the world
- Solving these “simple tasks” with traditional methods requires **a priori knowledge**, i.e. application-specific constraints that allow for a formal description of a **constrained sub-problem**
- As soon as the **assumptions** that underlie the formal description are **violated**, the system **fails** since its world model is no longer aligned with reality

Example: The Robotics Paradox



vs.



Robots excel at **well-defined repeatable tasks** where they can outperform even the most **skilled human workers**.

No robot built up to now can operate in **dynamic real-world environments** and carry out simple everyday tasks, such as walking, shopping, etc.

The Robotics Paradox – Factory vs. Real Life

Factory Task: Spot Welding

- Wait for the signal indicating that the two parts of the car body are in place
- Move the end-effector to pre-programmed position P1
- Activate the welding tool
- Make a spot weld by moving end-effector to P2
- Return to the resting position P0 after time t

Everyday Task: Go Shopping

- Write a shopping list
- Got to the shop
- Buy all items on the list
- Return home

The Robotics Paradox – Factory vs. Real Life

The welding task is well-defined because...

- ... the process of making a spot weld can be precisely described in terms of end-effector position and duration
- ...the complete environment of the robot is known and usually secured by a cage
- ... the robot behaves fully deterministically
- ... the environment behaves fully deterministically

The shopping task is ill-defined because...

- ... writing the shopping list depends on many external factors (appetite, special offers, menu, availability)
- ... going to the shop is different every time (time of day, traffic)
- ... the positions of the items in the shop might change without notice
- ... the availability of items has direct influence on the menu, which in turn affects the shopping list
- ... etc.

Cognitive System - Definition

Definition 2.1: Cognitive System

A cognitive system is an autonomous system that can perceive its environment, learn from experience, anticipate the outcome of events, act to pursue goals, and adapt to changing circumstances.

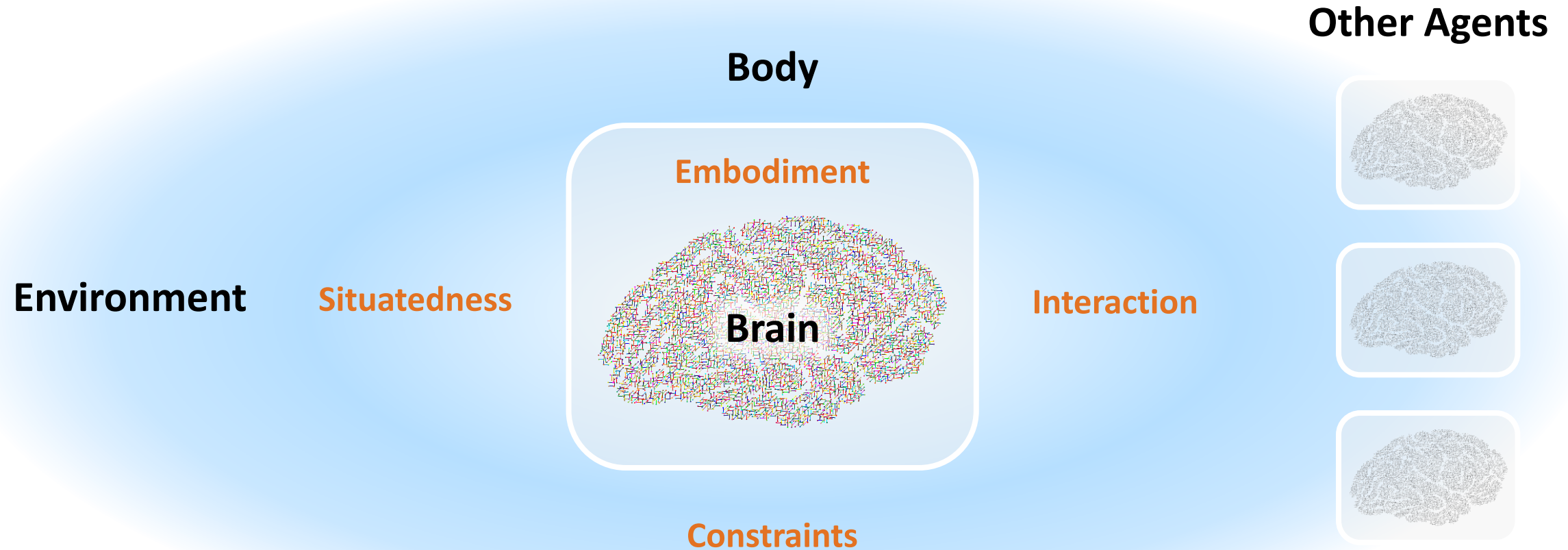
Vernon (2014)

Relation to Intelligence and Learning

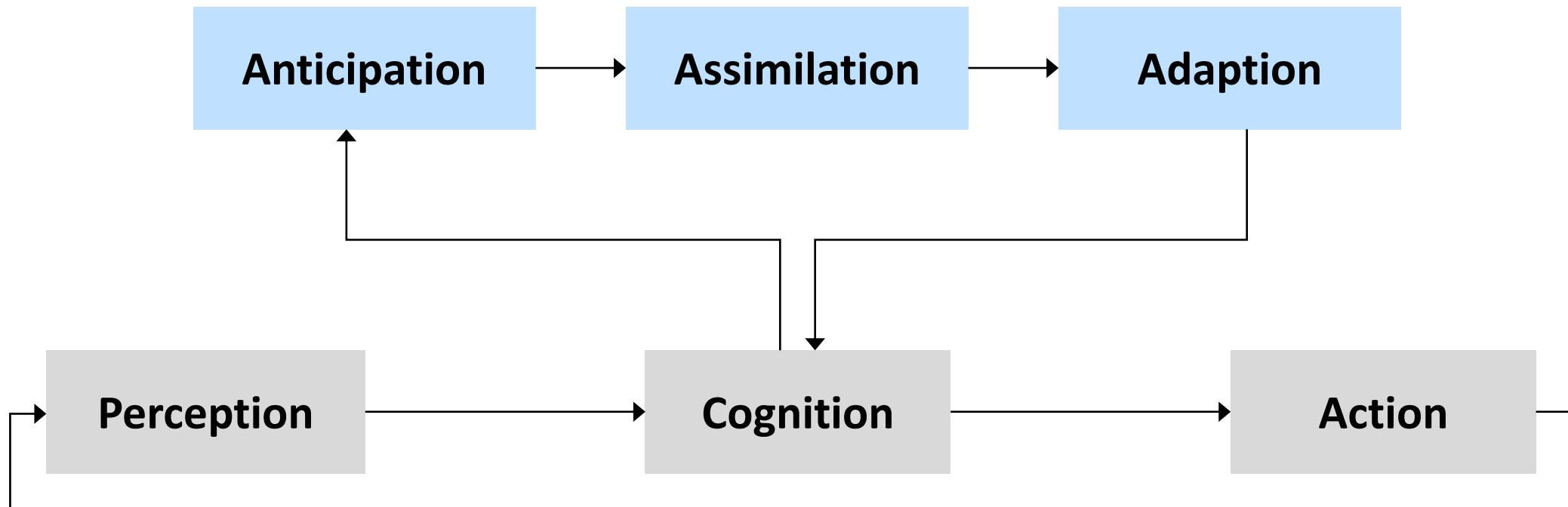
Cognition as a Synergy of Cognitive Skills

- Cognition is a **global process** at the **system level** that integrates many different processing modalities
- Special cognitive skills such as **intelligence**, **learning**, **memory**, **interaction**, etc. are **constituents** and **synergies** of a cognitive system
- The theory of cognitive systems provides a coherent **conceptual framework** that accommodates many other disciplines, such as **artificial intelligence**, **machine learning**, **computer vision**, **control theory**, etc.
- Cognitive systems research is therefore by definition a highly **interdisciplinary** field of research ranging from neuroscience to computer science and robotics
- In this course we will refer to the individual aspects of a cognitive system as **cognitive functions** or **cognitive skills**

Components of a Cognitive System



The Cycle of Cognitive Processing



Example Applications that Require Cognition

- Autonomous Driving
- Natural Language Processing
- Human-Robot Interaction
- Physical Interaction (e.g. walking, sports)
- Collaborative Work with Multiple Agents
- ...

Cognitive Capabilities: Self-Reliance

A cognitive system must be able to act in and interact with its environment self-reliantly, purposefully, and independently:

Goal-Directedness

Take on goals, formulate predictive strategies to achieve them, and put those strategies into effect

Autonomy

Operate with varying degrees of autonomy

Interaction

Interact – cooperate, collaborate, communicate – with other agents

From Vernon (2014)

Cognitive Capabilities: Perception & Action

A cognitive system must be able to perceive its environment, make sense of its perceptions, and predict future events:

Interpretation

Read the intention of other agents and anticipate their actions

Sensing

Sense and interpret expected and unexpected events

Anticipation

Anticipate the need for actions and predict the outcome of its own actions and those of others

Action

Select a course of action, carry it out, and then assess the outcome

From Vernon (2014)

Cognitive Capabilities: Adaption

A cognitive system must be able to adapt to changes of itself, of others, and within the environment:

Reaction

Adapt to changing circumstances, in real-time, by adjusting current and anticipated actions

Learning

Learn from experience: adjust the way actions are selected and performed in the future

Anomaly Detection

Notice when performance is degrading, identify the reason for the degradation, and take corrective action

From Vernon (2014)